

In-Class Worksheet - ANSWER KEY

Probability and Conditional Probability

Course: H524 Introduction to Biostatistics

Topic: Probability, Conditional Probability, and Diagnostic Testing

1 Problem 1: Mammogram Screening (Based on Gigerenzer's Study)

A 45-year-old woman with no symptoms or family history of breast cancer has a positive mammogram. What is the probability that she actually has breast cancer?

Given information:

- The prevalence of breast cancer in this age group is 1.2%
- If a woman has breast cancer, the probability is 85% that she will have a positive mammogram (sensitivity)
- If a woman does not have breast cancer, the probability is 9% that she will still have a positive mammogram (false positive rate)

Part A: Set up a 2x2 table assuming 10,000 women are screened. Fill in all cells:

	Cancer +	Cancer -	Total
Test +	102	889	991
Test -	18	8,991	9,009
Total	120	9,880	10,000

Work space for calculations:

- Total with cancer: $10,000 \times 0.012 = 120$
- Total without cancer: $10,000 - 120 = 9,880$
- True positives: $120 \times 0.85 = 102$
- False negatives: $120 - 102 = 18$
- False positives: $9,880 \times 0.09 = 889$
- True negatives: $9,880 - 889 = 8,991$

Part B: Calculate the Positive Predictive Value (PPV): $\Pr(\text{Cancer} \mid \text{Test}+)$

$$\text{PPV} = \frac{102}{991} = 0.103 = \mathbf{10.3\%}$$

Part C: Interpret your result. Why is the PPV so much lower than the sensitivity?

Answer: Even though the test has 85% sensitivity, only 10.3% of positive tests indicate actual cancer. This is because the prevalence is low (1.2%), so false positives (889) vastly outnumber true

positives (102). In a low-prevalence population, even a small false positive rate (9%) generates many false positives.

2 Problem 2: Strep Throat Testing

A rapid strep test has 85% sensitivity and 95% specificity. Among patients presenting with sore throat, 15% actually have strep throat.

Part A: Create a 2x2 table for 1,000 patients with sore throat:

	Strep +	Strep -	Total
Test +	128	43	171
Test -	22	807	829
Total	150	850	1,000

Calculations:

- Strep positive: $1,000 \times 0.15 = 150$
- Strep negative: $1,000 - 150 = 850$
- True positives: $150 \times 0.85 = 128$ (rounded)
- False negatives: $150 - 128 = 22$
- True negatives: $850 \times 0.95 = 808$ (rounded to 807 for total)
- False positives: $850 - 807 = 43$

Part B: Calculate the Negative Predictive Value (NPV): $\Pr(\text{No Strep} \mid \text{Test-})$

$$\text{NPV} = \frac{807}{829} = 0.973 = \mathbf{97.3\%}$$

Part C: What is the probability that a patient with a negative test actually has strep (false negative)?

$$\Pr(\text{Strep} \mid \text{Test-}) = \frac{22}{829} = 0.027 = \mathbf{2.7\%}$$

3 Problem 3: Smoking and Heart Disease

In a study of 1,000 people, there were 200 smokers and 800 non-smokers. During the 10-year follow-up, 30 smokers and 20 non-smokers developed heart disease.

Part A: Create a 2x2 table:

	Heart Disease	No Disease	Total
Smokers	30	170	200
Non-smokers	20	780	800
Total	50	950	1,000

Part B: Calculate the risk of heart disease in smokers:

$$\text{Risk}_{\text{smokers}} = \frac{30}{200} = 0.15 = \mathbf{15\%}$$

Part C: Calculate the risk of heart disease in non-smokers:

$$\text{Risk}_{\text{non-smokers}} = \frac{20}{800} = 0.025 = \mathbf{2.5\%}$$

Part D: Calculate the Relative Risk (RR):

$$\text{RR} = \frac{\text{Risk}_{\text{smokers}}}{\text{Risk}_{\text{non-smokers}}} = \frac{0.15}{0.025} = \mathbf{6.0}$$

Part E: Interpret the relative risk in plain language:

Answer: Smokers have 6 times the risk of developing heart disease compared to non-smokers over the 10-year period. Smoking is strongly associated with increased heart disease risk.

4 Problem 4: Binomial Probability

A new treatment has a 60% success rate.

Part A: If 10 patients receive the treatment, what is the probability that at least 8 of them respond successfully? Use R or a binomial probability table.

Using R:

```
pbinom(7, size = 10, prob = 0.6, lower.tail = FALSE)
```

$$\Pr(X \geq 8) = \Pr(X = 8) + \Pr(X = 9) + \Pr(X = 10) = \mathbf{0.167 \text{ or } 16.7\%}$$

Part B: What is the expected number of successful responses out of 10 patients?

$$E(X) = n \times p = 10 \times 0.6 = \mathbf{6 \text{ patients}}$$

5 Problem 5: Normal Distribution

Cholesterol levels in adults are normally distributed with mean 200 mg/dL and standard deviation 40 mg/dL.

Part A: What proportion of adults have cholesterol levels between 180 and 220 mg/dL?

Solution:

- $Z_1 = \frac{180-200}{40} = -0.5$
- $Z_2 = \frac{220-200}{40} = 0.5$
- $\Pr(-0.5 < Z < 0.5) = \Pr(Z < 0.5) - \Pr(Z < -0.5) = 0.6915 - 0.3085 = \mathbf{0.383 \text{ or } 38.3\%}$

Using R:

```
pnorm(220, mean = 200, sd = 40) - pnorm(180, mean = 200, sd = 40)
```

Part B: What cholesterol level corresponds to the 90th percentile?

Solution:

- The 90th percentile has $Z = 1.28$ (from Z-table)
- $X = \mu + Z\sigma = 200 + 1.28(40) = 200 + 51.2 = \mathbf{251.2 \text{ mg/dL}}$

Using R:

```
qnorm(0.90, mean = 200, sd = 40)
```